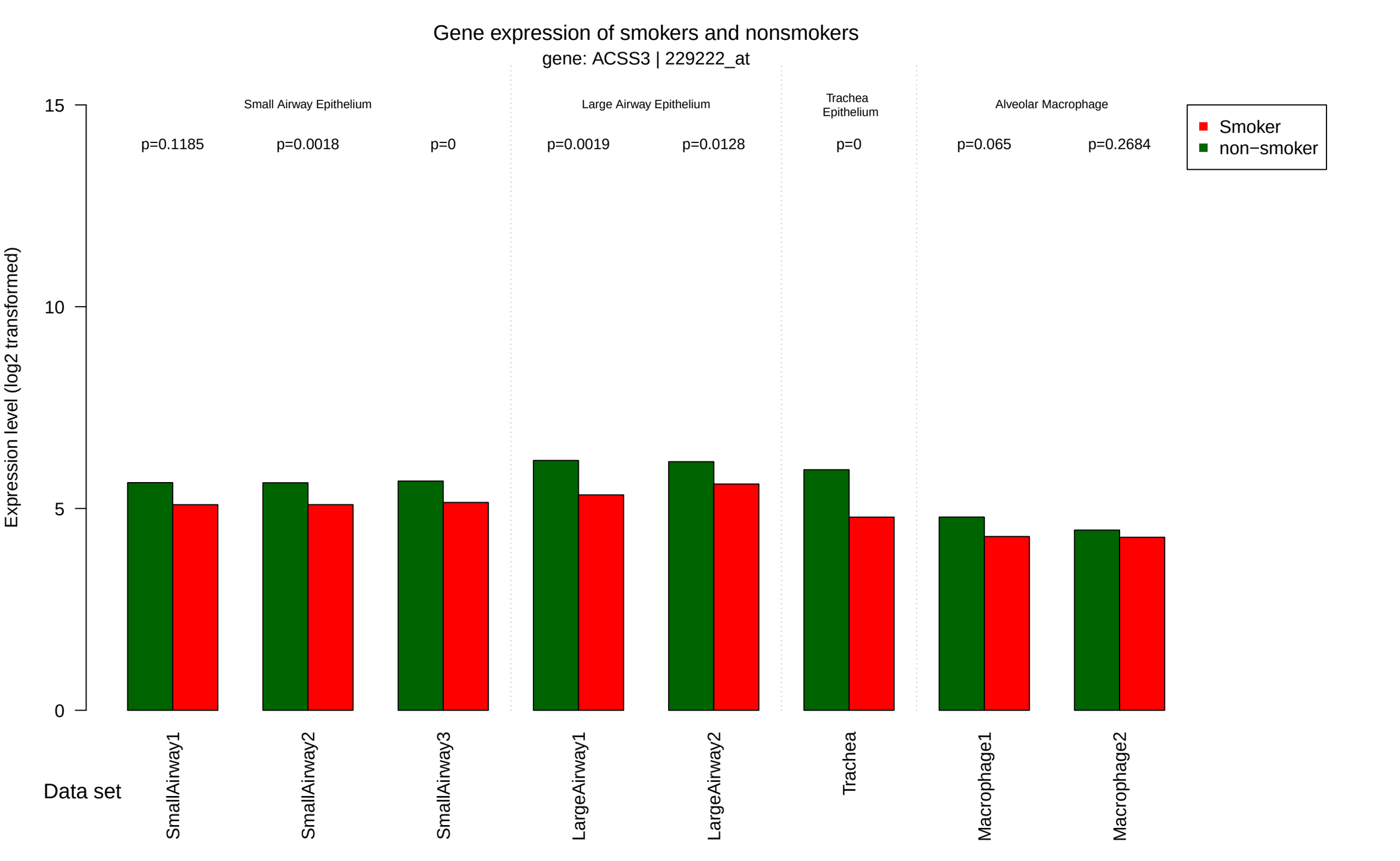
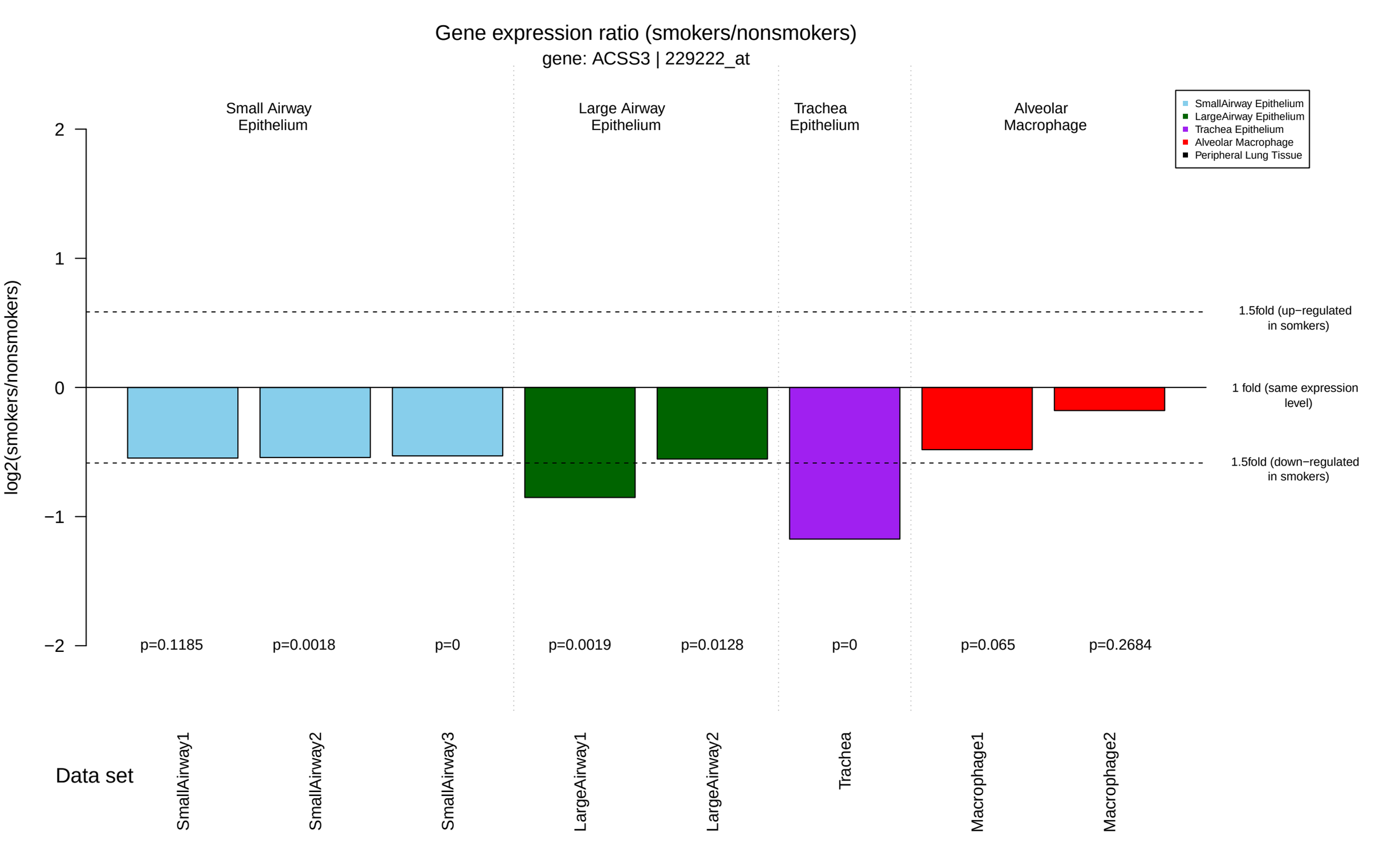
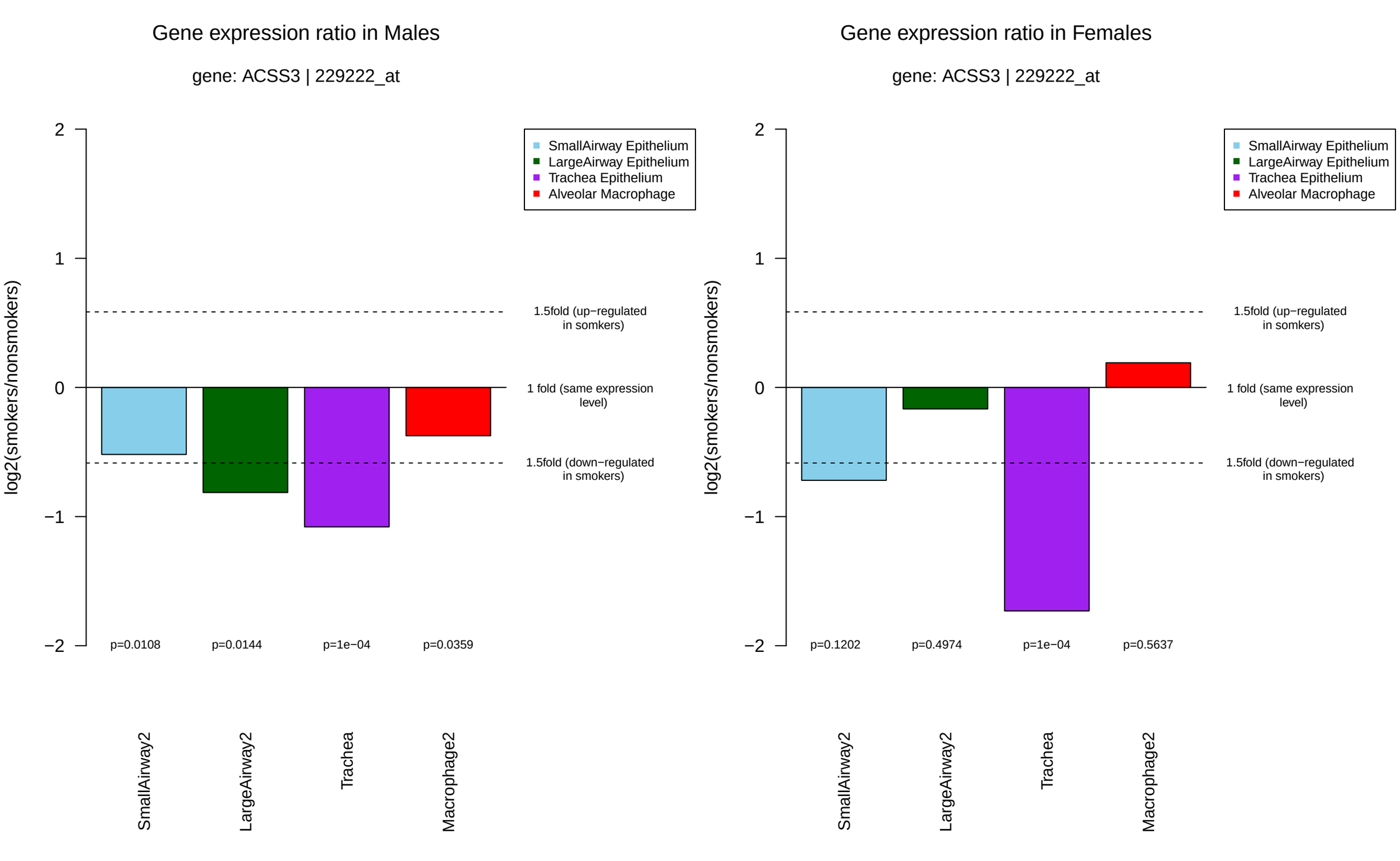
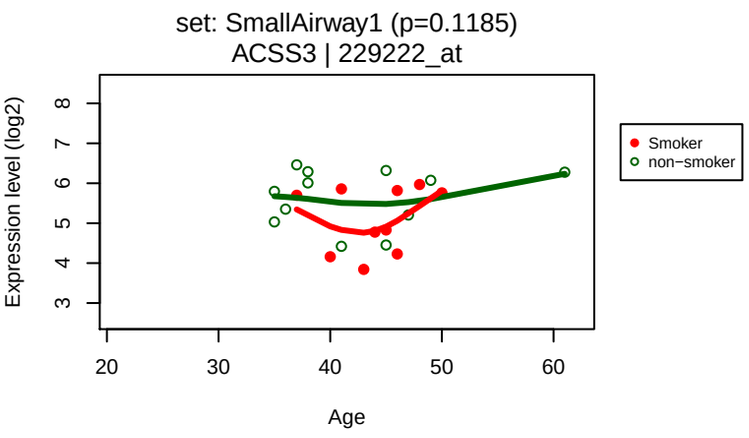
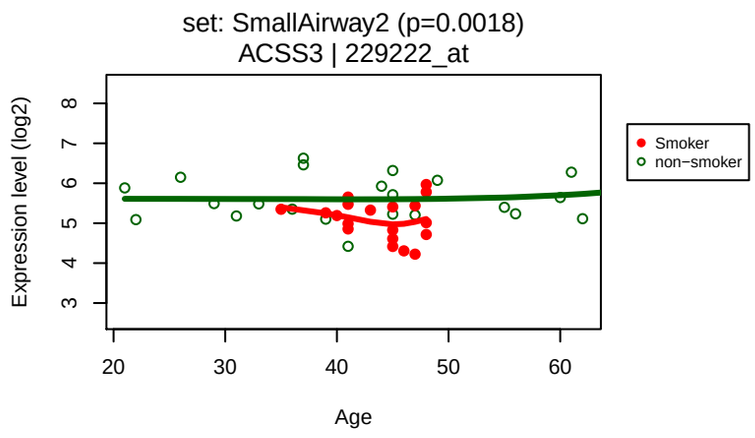
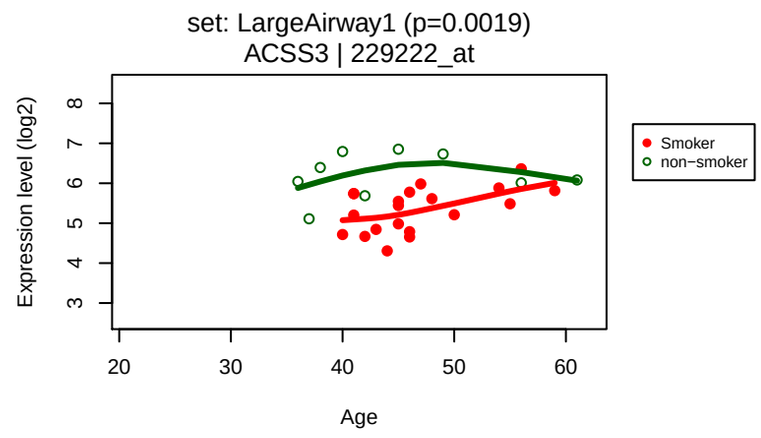

Gene expression of smokers and nonsmokers
gene: ACSS3 | 229222_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1185
p=0.0018
p=0
p=0.0019
p=0.0128
p=0
p=0.065
p=0.2684
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

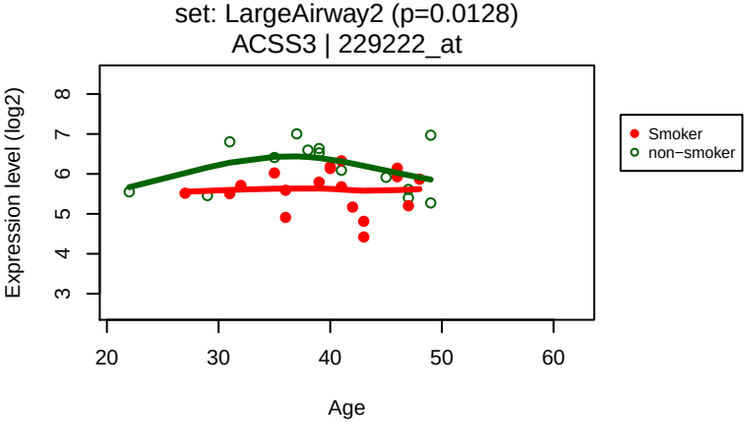

Gene expression ratio (smokers/nonsmokers)
gene: ACSS3 | 229222_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1185
p=0.0018
p=0
p=0.0019
p=0.0128
p=0
p=0.065
p=0.2684
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

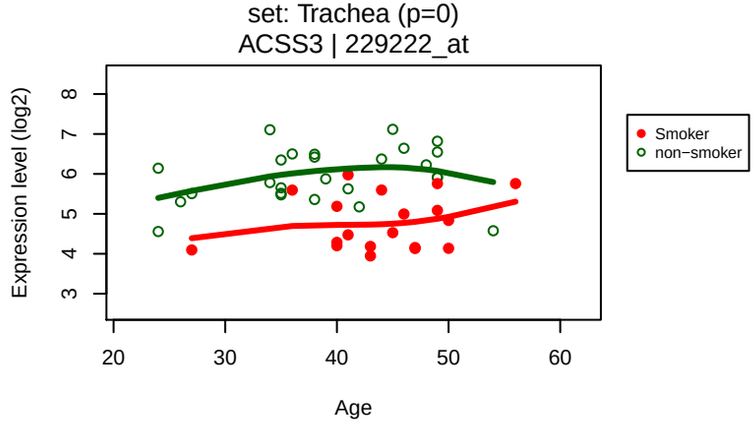

Gene expression ratio in Males
gene: ACSS3 | 229222_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0108
p=0.0144
p=1e−04
p=0.0359
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACSS3 | 229222_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1202
p=0.4974
p=1e−04
p=0.5637
SmallAirway2
LargeAirway2
Trachea
Macrophage2

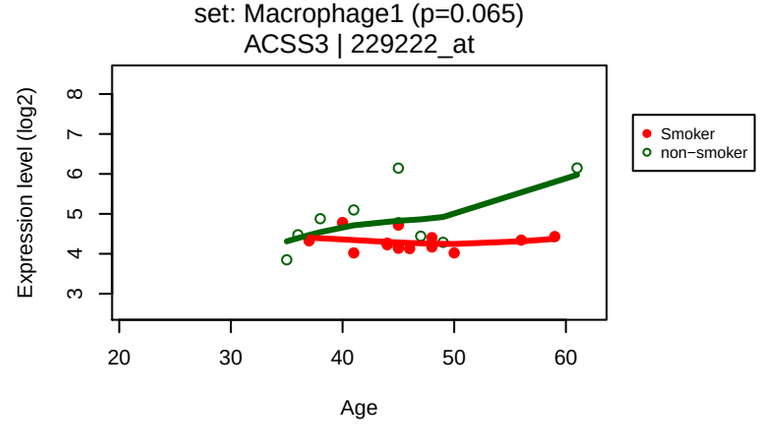

set: SmallAirway1 (p=0.1185)
ACSS3 | 229222_at
Expression level (log2)
Age
Smoker
non−smoker

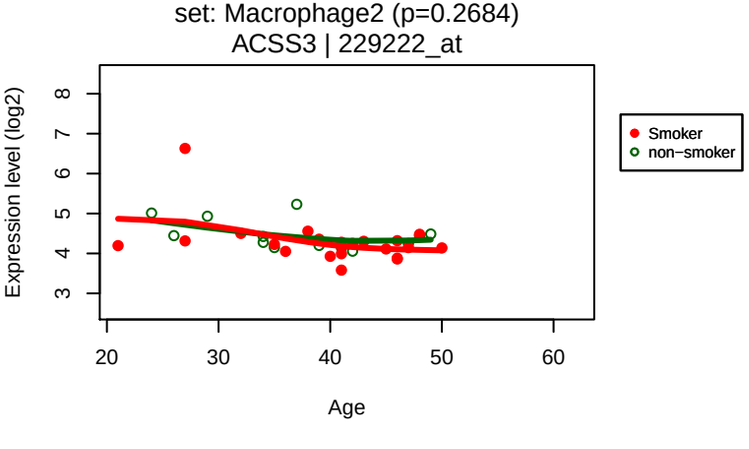

set: SmallAirway2 (p=0.0018)
ACSS3 | 229222_at
Expression level (log2)
Age
Smoker
non−smoker

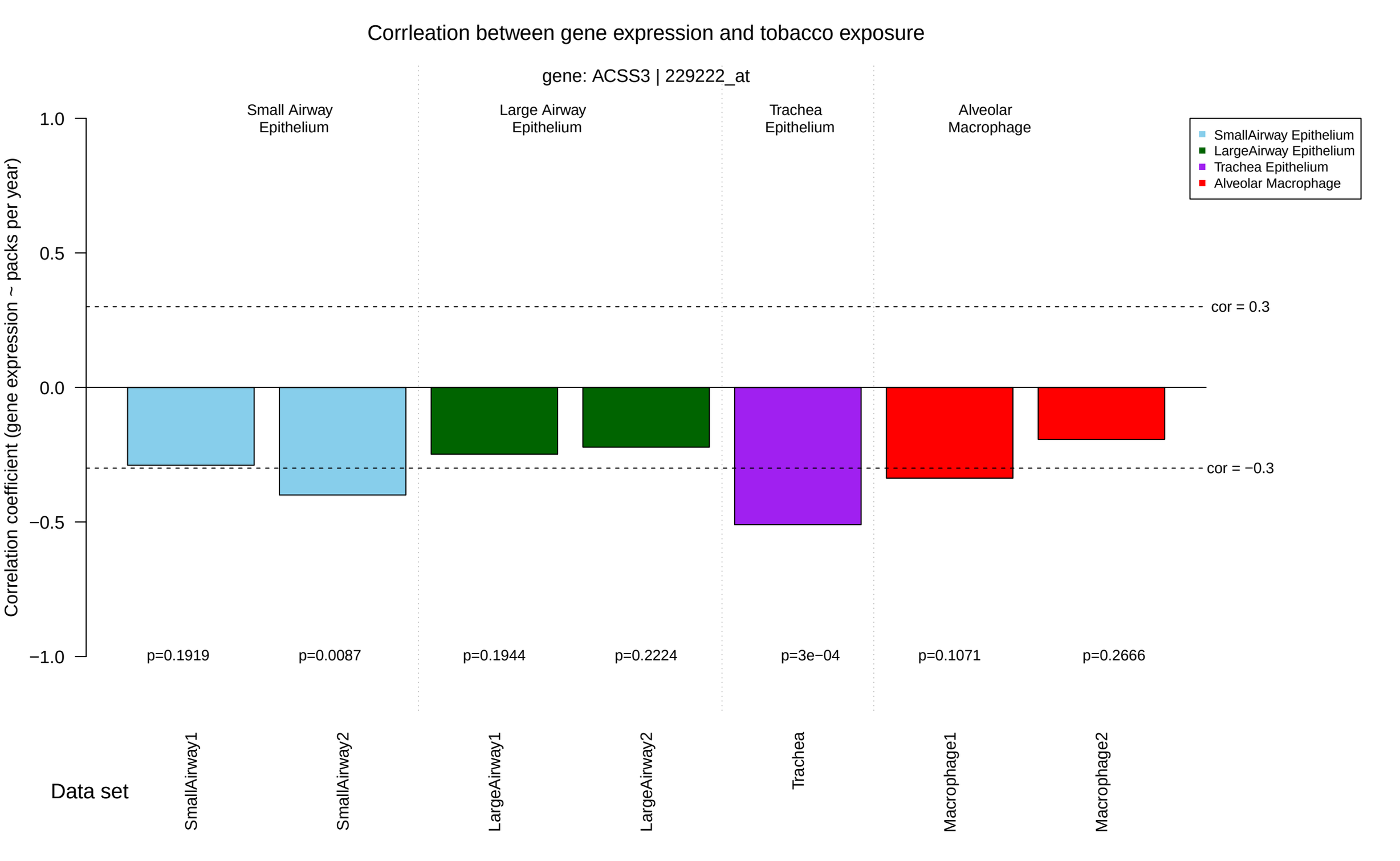

set: LargeAirway1 (p=0.0019)
ACSS3 | 229222_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0128)
ACSS3 | 229222_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
ACSS3 | 229222_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.065)
ACSS3 | 229222_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2684)
ACSS3 | 229222_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACSS3 | 229222_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1919
p=0.0087
p=0.1944
p=0.2224
p=3e−04
p=0.1071
p=0.2666
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2